1 Methodology of the Factor Correlation Analysis

1.1 Objective and Sample Design

The notebook evaluates the correlation structure, redundancy, stability, and economic relevance of a library of candidate factors for an AI-related equity basket. The analysis is divided into two periods:

- Estimation period: 2017-01-03 to 2023-12-29.
- Holdout period: 2024-01-02 to 2026-10-01.

All factor screening, correlation analysis, clustering, VIF diagnostics, and draft selection are performed using the estimation period only. The holdout period is used to assess stability and out-of-sample explanatory power.

1.2 Candidate Factor Construction

The candidate library contains market, rates, credit, volatility, foreign exchange, commodities, style, semiconductor, software, power, grid, uranium, cyber-security, and robotics factors.

The market factor is defined as the excess return of SPY:

$$MKT_t = R_t^{SPY} - R_t^f.$$

ETF-based sector and style factors are generally constructed as spreads relative to SPY. For example:

$$SEMI_t = R_t^{SMH} - R_t^{SPY},$$

$$SOFT_t = R_t^{IGV} - R_t^{SPY},$$

$$GRID_t = R_t^{GRID} - R_t^{SPY}.$$

Other examples include:

$$HY_t = R_t^{HYG} - R_t^{IEF},$$

$$POWER_t = R_t^{UTES} - R_t^{XLU},$$

$$URANIUM_t = R_t^{URA} - R_t^{SPY}.$$

Interest-rate factors are measured as changes in yield levels rather than percentage returns. For example, the 10-year Treasury factor is:

$$D10Y_t = 100 \times \left(y_t^{10Y} - y_{t-1}^{10Y}\right),$$

where the unit is basis points. Volatility is represented by the log change in the VIX.

1.3 Factor Eligibility Screening

A factor is retained only when it satisfies the following conditions in the estimation window:

- it has data from the beginning of the estimation period;
- it has at least 98% valid observations;
- its ETF legs pass the liquidity and data-quality filters from the cleaning notebook.

This excludes short-history factors that cannot support a stable long-window analysis. The screen ensures that later correlation and regression results are not driven by different sample lengths.

1.4 Market Neutralisation

Many factors are partly driven by broad equity-market movements. To isolate their distinct contribution, every non-market factor is market-neutralised using its estimation-period beta:

$$\beta_i = \frac{\mathrm{Cov}(F_i, MKT)}{\mathrm{Var}(MKT)}.$$

The market-neutral version of factor i is:

$$F_i^{\perp} = F_i - \beta_i MKT.$$

This transformation removes the part of each factor that is linearly related to the market. Correlations between market-neutral factors are therefore more informative than raw correlations when assessing whether factors contain distinct information.

1.5 Correlation and Robustness Analysis

The notebook computes three correlation measures for the eligible factor set:

- Pearson correlation;
- Spearman rank correlation;
- Pearson correlation after robust winsorisation.

Winsorisation limits extreme observations using the median and median absolute deviation:

$$F_{i,t}^{W} = \min\left[\max\left(F_{i,t}, \mathrm{median}(F_i) - k \cdot 1.4826 \cdot MAD(F_i)\right), \mathrm{median}(F_i) + k \cdot 1.4826 \cdot MAD(F_i)\right].$$

Comparing ordinary, rank-based, and winsorised correlations tests whether the apparent relationship between two factors is robust or mainly driven by extreme market events.

The notebook also computes one-day lead-lag correlations. This is a diagnostic for stale prices, asynchronous market closes, or timing effects in non-U.S. instruments.

1.6 Clustering and Multicollinearity Diagnostics

Factors are grouped using hierarchical average-linkage clustering. The clustering distance is based on the absolute market-neutral correlation:

$$d_{ij} = \sqrt{\frac{1 - |\rho_{ij}|}{2}}.$$

Both strongly positive and strongly negative correlations are treated as redundant because either relationship can create multicollinearity in a regression.

For each factor, the notebook calculates the variance inflation factor:

$$VIF_i = \left(R^{-1}\right)_{ii},$$

where R is the factor correlation matrix. A VIF above 5 is treated as a warning, while a value above 10 indicates substantial multicollinearity.

The notebook also reports:

- the condition number of the correlation matrix;
- eigenvalues and cumulative explained variance;
- the effective number of independent factors.

These diagnostics identify whether a proposed factor set is suitable for regression or needs further reduction.

1.7 Stability and Regime Analysis

The notebook estimates rolling 252-trading-day market-neutral correlations. Rolling estimates are compared with the full estimation-window correlation matrix. This shows whether relationships between factors remain stable over time.

The analysis also compares factor structure across several regimes:

- low-, medium-, and high-VIX environments;
- the COVID-19 market shock;
- the 2022 rates shock;
- the 2025 stress episode;
- the full holdout period.

For each regime, the notebook reports average absolute correlation, the effective number of independent factors, and the condition number. This identifies periods in which diversification between factors becomes weaker.

1.8 AI Basket Construction

The AI basket is divided into economically meaningful layers:

- compute and semiconductors;
- networking and servers;
- hyperscalers and software;
- data centres and neoclouds;
- power generation;
- grid and electrical equipment;
- nuclear and uranium.

Within each layer, returns are equal-weighted across available securities. The basket also includes an aggregate version and a core version containing names with sufficient history during the estimation period.

The notebook estimates the market beta and market R^2 of each layer separately in the estimation and holdout periods. This reveals whether the AI basket's market sensitivity changed over time.

1.9 Factor Relevance to the AI Basket

For each basket layer and each market-neutral factor, the notebook estimates correlation and Newey–West adjusted significance statistics. The relevant regression is:

$$R_t^{basket} = \alpha + \gamma F_t^{\perp} + \varepsilon_t.$$

Newey–West standard errors are used because daily financial returns may exhibit heteroskedasticity and autocorrelation.

The analysis also controls for ETF holdings overlap. For example, a semiconductor ETF may hold many compute-layer companies. The notebook therefore compares factor relevance for:

- the full AI basket;
- the basket excluding layers that overlap with the factor ETF holdings.

This distinction helps separate a genuine economic relationship from mechanical overlap between ETF constituents and basket constituents.

1.10 AI-Closeness Classification

Factors are classified into three groups:

- distant factors, such as rates, credit, the dollar, commodities, and generic style;
- adjacent factors, such as power, grid, copper, and uranium;
- AI-close factors, such as semiconductors, software, cloud infrastructure, and robotics.

The classification is checked empirically. A factor is labelled empirically AI-close when its market-neutral correlation with a non-overlapping AI basket layer exceeds a specified threshold. This prevents a factor that appears generic in theory, such as momentum, from being treated as independent if it has become strongly related to AI equities in the data.

1.11 Nested Model Sets and Staged Regressions

The notebook compares nested factor sets:

- market only;
- distant factors;
- distant plus adjacent factors;
- all factors, including AI-close factors.

It also estimates a factor ladder. The stages are:

1. market only;
2. market plus macro factors;
3. market plus macro and style factors;
4. market plus adjacent power, grid, and uranium factors;
5. market plus all previous factors and AI-close factors.

For each stage, the notebook reports:

- in-sample R^2;
- holdout R^2 using coefficients estimated in the estimation period;
- holdout R^2 after re-fitting coefficients in the holdout;
- incremental R^2 relative to the previous stage.

The fixed-coefficient holdout result is the most important measure because it tests whether the earlier estimated relationship remains useful in the later period.

1.12 Bootstrap Inference

Confidence intervals for staged R^2 values and incremental explanatory power are estimated using a circular block bootstrap. Blocks of consecutive trading days are resampled rather than individual days. This preserves part of the time-series dependence in financial returns.

An incremental factor block is considered meaningful only when:

- its bootstrap confidence interval excludes zero; and
- the absolute increase in R^2 exceeds a minimum economic threshold.

1.13 Interpretation

The notebook is designed as a factor-selection and diagnostic stage, not as the final attribution model. Its main purpose is to identify a compact, interpretable, and sufficiently independent factor set for later orthogonalisation, rolling regression, and return and risk attribution.